

Solution 1
System Theory Basics

Exercise 1 Equilibrium Point and Linearization

1) The resulting force $F(t)$ is:

$$F(t) = ka(\sin \theta_2(t) - \sin \theta_1(t)) + da(\cos(\theta_2(t))\dot{\theta}_2(t) - \cos(\theta_1(t))\dot{\theta}_1(t)).$$

2) One trivial equilibrium is given by the origin. Because of symmetry we have in addition for $\theta_1 = -\theta_2$ that

$$0 = mgl \sin \theta_1 + a \cos \theta_1 (ka(-2\sin \theta_1)) \quad (1)$$

$$\Leftrightarrow \cos(\theta_1) = \frac{mgl}{2a^2k} \quad (2)$$

yields two additional equilibrium points, which are given by $\theta_2 = -\theta_1$ and $\theta_1 = \pm \arccos \frac{mgl}{2a^2k}$ since $\frac{mgl}{2a^2k} \leq 1$.

In the second case it follows from $\dot{\theta}_1(t) = \dot{\theta}_2(t) = \ddot{\theta}_1(t) = \ddot{\theta}_2(t) = 0$, and $\bar{\theta}_2 = 0$ that

$$\begin{aligned} mgl \sin \theta_1(t) - a^2k \cos \theta_1(t) \sin \theta_1(t) + T(t) &= 0 \\ a^2k \sin \theta_1(t) + \alpha &= 0. \end{aligned}$$

Hence,

$$\begin{aligned} \bar{\theta}_1 &= \arcsin \left(-\frac{\alpha}{a^2k} \right) \\ \bar{T} &= mgl \frac{\alpha}{a^2k} - \alpha \cos \arcsin \left(-\frac{\alpha}{a^2k} \right). \end{aligned}$$

Note, that we neglected angles $|\bar{\theta}_1|$ that are larger than $\pi/2$ due to the simplifying assumption that vertical movement can be ignored.

3) The new equilibrium point is $\bar{\theta}_1 = \bar{\theta}_2 = \bar{T} = 0$. We set the state and input variables as follows

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} \theta_1(t) - \bar{\theta}_1 = \theta_1(t) \\ \dot{\theta}_1(t) \\ \theta_2(t) - \bar{\theta}_2 = \theta_2(t) \\ \dot{\theta}_2(t) \end{bmatrix}, \quad u(t) = T(t) - \bar{T} = T(t)$$

The system of four 1st order ODEs $\dot{x}(t) = f(x(t), u(t))$ is given by

$$f(x(t), u(t)) =$$

$$\begin{bmatrix} x_2(t) \\ \frac{1}{m l^2} (mgl \sin x_1(t) + a^2 \cos x_1(t) (k(\sin x_3(t) - \sin x_1(t)) + d(\cos(x_3(t))x_4(t) - \cos(x_1(t))x_2(t))) + u(t)) \\ x_4(t) \\ \frac{1}{m l^2} (mgl \sin x_3(t) - a^2 \cos x_3(t) (k(\sin x_3(t) - \sin x_1(t)) + d(\cos(x_3(t))x_4(t) - \cos(x_1(t))x_2(t))) + \alpha + \beta x_4^2(t)) \end{bmatrix}$$

Then linearizing with

$$A = \left. \frac{\partial f(x(t), u(t))}{\partial x(t)} \right|_{x(t)=\bar{x}, u(t)=\bar{u}}, \quad B = \left. \frac{\partial f(x(t), u(t))}{\partial u(t)} \right|_{x(t)=\bar{x}, u(t)=\bar{u}}$$

leads to a state space description of the form $\dot{x}(t) = Ax(t) + Bu(t)$ given by

$$\dot{x}(t) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{g}{l} - \frac{a^2 k}{ml^2} & -\frac{a^2 d}{ml^2} & \frac{a^2 k}{ml^2} & \frac{a^2 d}{ml^2} \\ 0 & 0 & 0 & 1 \\ \frac{a^2 k}{ml^2} & \frac{a^2 d}{ml^2} & \frac{g}{l} - \frac{a^2 k}{ml^2} & -\frac{a^2 d}{ml^2} \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ \frac{1}{ml^2} \\ 0 \\ 0 \end{pmatrix} u(t)$$

$$y(t) = (1 \ 0 \ 0 \ 0) x(t) + 0 \cdot u(t).$$

Exercise 2 Discretization of a LTI continuous-time state-space model

1) Euler's first order approximation is

$$\dot{x}(t) \approx \frac{x(t + T_s) - x(t)}{T_s}$$

and thus

$$x(k + 1) = x(k) + T_s Ax(k) + T_s Bu(k).$$

The resulting discrete-time matrices are

$$A^d = I + T_s A = \begin{bmatrix} -1.5 & 1.35 \\ -1.55 & 1.75 \end{bmatrix}, \quad B^d = T_s B = \begin{bmatrix} 2 & 1.05 \\ 0.55 & 1.5 \end{bmatrix},$$

$$C^d = C^c = [1 \ 1], \quad D^d = D^c = [0 \ 0].$$

2) The discrete-time matrices A^d and B^d can be calculated as

$$x(t_{k+1}) = \underbrace{e^{A^c T_s}}_{A^d} x(t_k) + \underbrace{\int_{\tau=0}^{T_s} e^{A^c(T_s-\tau)} B^c d\tau}_{B^d} u(t_k)$$

with the matrix exponential

$$A^d = e^{A^c T_s} = \sum_{k=0}^{\infty} \frac{(A^c T_s)^k}{k!}$$

and

$$\int_{\tau=0}^{T_s} e^{A^c(T_s-\tau)} B^c d\tau = \left[-(A^c)^{-1} e^{A^c(T_s-\tau)} B^c \right]_{\tau=0}^{T_s}$$

$$= -(A^c)^{-1} (e^0 - e^{A^c T_s}) B^c = (A^c)^{-1} (A^d - I) B^c$$

if A^c is regular. The system can be discretized with the Matlab command $A^d = \text{expm}(AT_s)$. The resulting discrete-time matrices are

$$A^d = \begin{bmatrix} -0.2046 & 0.6156 \\ -0.7068 & 1.277 \end{bmatrix}, \quad B^d = \begin{bmatrix} 0.6871 & 0.8462 \\ -0.2614 & 1.314 \end{bmatrix}, \quad C^d = C^c = [1 \ 1], \quad D^d = D^c = [0 \ 0].$$

Remark (Relation to Euler's discretization): A first order approximation of the matrix exponential is

$$A^d = e^{A^c T_s} \approx \sum_{k=0}^1 \frac{1}{k!} (A^c T_s)^k = I + A^c T_s$$

and thus, a linear approximation is $A_d = I + A^c T_s$ and $B_d = B^c T_s$, which is the result of Euler's discretization in 1).

- 3) A step response of the continuous-time system and discrete-time system with Euler and exact discretization with initial state $x(0) = [0; 0]$ and constant input $u(t) = [1; 1]$ is given in Figure 1. The step response for the discrete-time system with Euler's discretization oscillates heavily, which is an indication that the sampling-time is chosen too high for the linear approximation of Euler's discretization. A sampling-time of $T_s = 0.05$ is more appropriate and reflects the system's behavior quite well.

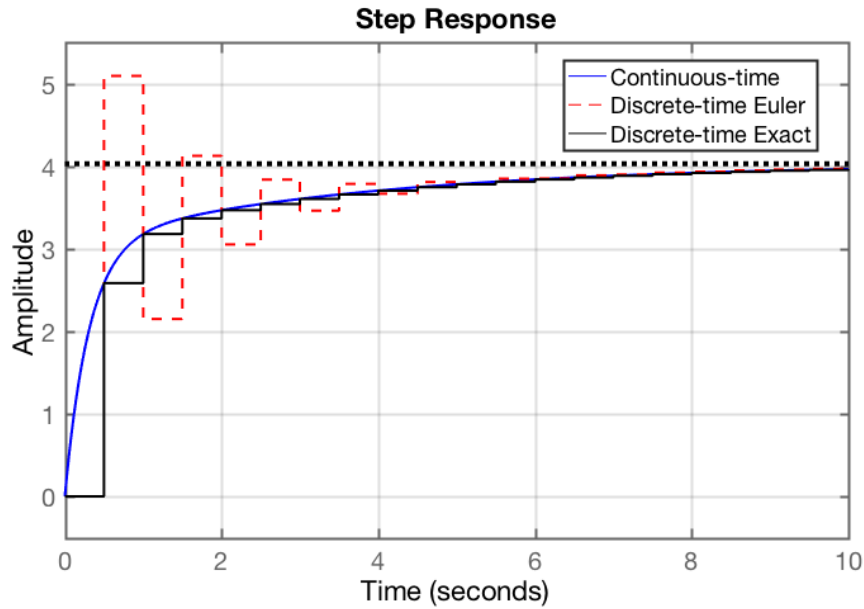


Figure 1: Comparison of continuous and discretized output in dynamic simulation.

Exercise 3 Sum of Lyapunov functions

$$\begin{aligned}
 V(x(k+1)) - V(x(k)) &= \alpha x^T(k) A^T P_1 A x(k) + (1 - \alpha) x^T(k) A^T P_2 A x(k) \\
 &\quad - \alpha x^T(k) P_1 x(k) - (1 - \alpha) x^T(k) P_2 x(k) \\
 &\leq -\alpha x^T(k) \Gamma x(k) - (1 - \alpha) x^T(k) \Gamma x(k) \\
 &= -x^T(k) \Gamma x(k)
 \end{aligned}$$

Exercise 4 Lyapunov Functions

The closed-loop system has the form $x(k+1) = A_{cl}x(k)$, where

$$A_{cl} = \begin{bmatrix} -0.4 & -1.1 & 0 \\ -0.4 & 0.55 & 0 \\ 0 & 0 & 0.9 \end{bmatrix}.$$

The Lyapunov candidate $V(x(k))$ is positive definite, since the matrix P is positive definite. It also satisfies $V(0) = 0$ and $V(x(k)) \rightarrow \infty$ when $\|x(k)\| \rightarrow \infty$. The latter property indicates that, if $x(k) = 0$ is asymptotically stable, it is globally asymptotically stable, too. The energy decrease equation yields:

$$V(A_{cl}x(k)) - V(x(k)) = x^T(k)(A_{cl}^T P A_{cl} - P)x(k) = x^T(k)Qx(k).$$

By substituting the values, one obtains:

$$Q = A_{cl}^T P A_{cl} - P = \begin{bmatrix} -0.52 & 0 & 0 \\ 0 & -0.185 & 0 \\ 0 & 0 & -0.19 \end{bmatrix}.$$

The eigenvalues of the matrix Q are -0.52 , -0.185 and -0.19 . All eigenvalues are negative and Q is symmetric. Therefore, Q is negative definite, which means that the function $V(x(k))$ is indeed a Lyapunov function for the closed-loop system. This implies that the closed-loop system is stable.